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Semiconductor structure and method for manufacturing same

Abstract

A method for manufacturing a semiconductor structure includes: providing a substrate; forming a gate trench in the substrate, the gate trench including a first trench and a second trench, the second trench being located above the first trench, communicating with the first trench, and having a width greater than a width of the first trench; and forming a gate word line in the gate trench.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/CN2021/106660 filed on Jul. 16, 2021, which claims priority to China Patent Application No. 202110351086.2 filed on Mar. 31, 2021. The disclosures of the above-referenced applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Dynamic storages are developing towards high speed, high integration density, low power consumption, etc. With the miniaturization of the structural size of semiconductor devices, the size of a gate word line is miniaturized, especially in a process for manufacturing Dynamic Random Access Memory (DRAM) with a key size less than 20 nm.

SUMMARY

(2) The disclosure relates generally to the technical field of integrated circuits, and more specifically to a semiconductor structure and a method for manufacturing the same.

- (3) According to some embodiments of the present disclosure, a semiconductor structure and a method for manufacturing the same are provided.
 - (4) A method for manufacturing a semiconductor structure includes the following operations.
 - (5) A substrate is provided.
 - (6) A gate trench is provided in the substrate. The gate trench includes a first trench and a second trench. The second trench is located above the first trench, communicates with the first trench, and has a width greater than a width of the first trench.
 - (7) A gate word line is formed in the gate trench.
 - (8) A semiconductor structure includes: a substrate; a gate trench located in the substrate and including a first trench and a second trench, and a gate word line located in the gate trench. The second trench is located above the first trench, communicated with the first trench, and has a width greater than a width of the first trench.
 - (9) The descriptions only summarize the technical solutions of this disclosure. To know the technical means of this disclosure more clearly, and implement according to the contents of the description, the following will describe by combining preferred embodiments of this disclosure and the brief description of the drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to describe the technical solutions in the embodiments of the present disclosure or in the related art more clearly, drawings required to be used in the embodiments or the illustration of the traditional technology will be briefly introduced below. Apparently, the drawings in the description below are only some examples of the disclosure. Those ordinarily skilled in the art also can acquire other drawings according to the provided drawings without creative work.
- (2) FIG. 1 shows a flowchart of a method for manufacturing a semiconductor structure provided by an embodiment of the disclosure.
- (3) FIG. 2 schematically shows the cross section of an array region of a structure obtained at S10 in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (4) FIG. 3 schematically shows the cross section of an array region of a structure obtained after a first trench is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (5) FIG. 4 schematically shows the cross section of an array region of a structure obtained after a first oxide layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (6) FIG. 5 schematically shows the cross section of an array region of a structure obtained after a sacrificial layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (7) FIG. 6 schematically shows a first cross section of an array region of a structure obtained after a sacrificial pattern is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (8) FIG. 7 schematically shows a second cross section of an array region of a structure obtained after a sacrificial pattern is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (9) FIG. 8 schematically shows the cross section of an array region of a structure obtained after a side wall material layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.
- (10) FIG. 9 schematically shows the cross section of an array region of a structure obtained after a

side wall pattern is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(11) FIG. **10** schematically shows the cross section of an array region of a structure obtained after an isolation groove is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(12) FIG. **11** schematically shows the cross section of an array region of a structure obtained after a filling material layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(13) FIG. **12** schematically shows the cross section of an array region of a structure obtained after a filling layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(14) FIG. **13A** schematically shows the cross section of an array region of a structure obtained after a filling layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(15) FIG. **13B** schematically shows the cross section of an edge region of a structure obtained after a filling layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(16) FIG. **14A** schematically shows the cross section of an array region of a structure obtained after a second oxide layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(17) FIG. **14B** schematically shows the cross section of an edge region of a structure obtained after a second oxide layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(18) FIG. **15A** schematically shows the cross section of an array region a structure obtained after a conductive material layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(19) FIG. **15B** schematically shows the cross section of an edge region a structure obtained after a conductive material layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(20) FIG. **16A** schematically shows the cross section of an array region of a structure obtained after a word line conductive layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(21) FIG. **16B** schematically shows the cross section of an edge region of a structure obtained after a word line conductive layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(22) FIG. **17A** schematically shows the cross section of an array region of a structure obtained after an auxiliary conductive layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(23) FIG. **17B** schematically shows the cross section of an edge region of a structure obtained after an auxiliary conductive layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(24) FIG. **18A** schematically shows the cross section of an array region of a structure obtained after a dielectric cover layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(25) FIG. **18B** schematically shows the cross section of an edge region of a structure obtained after a dielectric cover layer is formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

(26) FIG. **19** schematically shows the cross section of an array region of a structure obtained after a first interconnection structure and a second interconnection structure are formed in a method for manufacturing the semiconductor structure provided by an embodiment of the disclosure.

DETAILED DESCRIPTION

(27) For convenience of an understanding of the present disclosure, the embodiments of the disclosure will be described more fully below with reference to the related accompanying drawings. Preferred embodiments of the present disclosure are shown in the accompanying diagrams. The present disclosure may, however, be embodied in many different forms which are not limited to the embodiments described herein. Rather, these embodiments are provided so that the disclosed content of the present disclosure will be more thorough and complete.

(28) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which the present disclosure belongs. The terms used herein in the specification of the present disclosure are for the purpose of describing specific embodiments only and are not intended to be limiting of the present disclosure. The term “and/or” used herein covers any and all combinations of one or more related listed items.

(29) In the description of the embodiments of the present disclosure, it should be noted that orientations or positional relationships indicated by the terms “upper”, “lower”, “vertical”, “horizontal”, “inside”, “outside” and the like are orientations or positional relationships as shown in the drawings, and are only for the purpose of facilitating and simplifying the description of the embodiments of the present disclosure instead of indicating or implying that devices or elements indicated must have particular orientations, and be constructed and operated in the particular orientations. Therefore, these terms cannot be explained as limitation on the present disclosure.

(30) The quality of DRAM in aspects of electrical properties can be directly determined by a contact resistance between the gate word line and a conductive interconnecting structure. The magnitude of the current and the delay time of the signal are directly determined by the resistance of the connection. Especially, for the gate word line, the magnitude and stability of the current signal directly determine the storage speed and storage efficiency of a device.

(31) In some implementations, the gate word line has a relatively small width at the top. During the formation of an interconnecting through hole electrically connected to the gate word line, the offset of photolithography alignment or the offset in a pattern transfer process will cause the formed interconnecting structure to be shifted relative to the gate word line, resulting in a relatively small contact area, and a relatively high contact resistance between the interconnecting structure and the gate word line.

(32) Referring to FIG. 1, various embodiments of the present disclosure provide a method for manufacturing a semiconductor structure including the following operations.

(33) At **S10**, a substrate is provided.

(34) At **S20**, a gate trench is formed in the substrate. The gate trench includes a first trench and a second trench. The second trench is located above the first trench, communicates with the first trench, and has a width greater than a width of the first trench.

(35) At **S30**, a gate word line is formed in the gate trench.

(36) In the method for manufacturing the semiconductor structure of the embodiments of the present disclosure, the gate word line with a wider upper part can be obtained by preparing the gate trench with an upper part wider than a lower part. Since the gate word line has the wider upper part, it is convenient to align and contact the first interconnecting structure with the gate word line.

Therefore, it can ensure that a sufficiently large contact area can be formed between the first interconnection structure and the gate word line, thereby reducing the contact resistance between the first interconnection structure and the gate word line, and improving electrical properties of the storage device to endow the storage device with higher read and write speed and storage efficiency.

(37) Referring to **S10** in FIG. 1 and FIG. 2, the substrate **10** is provided at **S10**.

(38) As an example, the substrate **10** may include, but is not limited to, a silicon substrate. The substrate **10** includes an active area **101** and an isolation structure **102**.

(39) Referring to **S20** in FIG. 1 and FIG. 3 through FIG. 13B, a gate trench **103** is formed in the substrate **10** at **S20**. The gate trench **103** includes a first trench **1031** and a second trench **1032**. The

second trench **1032** is located above the first trench **1031**, communicates with the first trench **1031**, and has width greater than a width of the first trench **1031**.

(40) As an example, **S20** may include the following operations.

(41) At **S201**, the first trench **1031** is formed in the substrate **10**, as shown in FIG. 3.

(42) At **S202**, a sacrificial layer **11** is formed. The sacrificial layer **11** includes a first part **111** and a second part **112**. The first trench **1031** is filled up by the first part **111**. The second part **112** covers an upper surface of the substrate **10** and an upper surface of the first part **111**, as shown in FIG. 5.

(43) At **S203**, an isolation groove **113** is formed in the second part **112** to form a sacrificial pattern **1121** by patterning the second part **112**. The sacrificial pattern **1121** is arranged to correspond to the first part **111**, and has a width greater than a width of the first part **111**, as shown in FIG. 10.

(44) At **S204**, a filling layer **15** is formed in the isolation groove **113**. The isolation groove **113** is filled up by the filling layer **15**, as shown in FIG. 12.

(45) At **S205**, the sacrificial pattern **1121** is removed to form the second trench **1032**.

(46) At **S206**, the first part **111** is removed, as shown in FIG. 13A and FIG. 13B.

(47) As an example, the method may include the following operations between **S201** and **S202**. A first oxide layer **161** is formed at a side wall of the first trench **1031**, as shown in FIG. 4. Specifically, the first oxide layer **161** may be formed by using, but not limited to, a thermal oxidization process.

(48) As an example, the sacrificial layer **11** formed at **S202** may include, but is not limited to, a titanium nitride layer.

(49) As an example, at **S203**, self-aligned dual imaging patterning (SADP) may be used to form the isolation groove **113** in the second part **112**. Specifically, **S203** may include the following operations.

(50) At **S2031**, a mask layer **12** is formed on the sacrificial layer **11**. The mask layer **12** includes a first hard mask layer **121**, an organic sacrificial layer **122**, and a second hard mask layer **123** that are sequentially stacked from bottom to top. The first hard mask layer **121** may include, but is not limited to, a silicon oxynitride layer. The organic sacrificial layer **122** may include, but is not limited to, a spin-coated organic carbon layer. The second hard mask layer **123** may include, but is not limited to, a silicon oxynitride layer.

(51) At **S2032**, a patterned photoresist layer **13** is formed on the mask layer **12**. The patterned photoresist layer **13** delimits a position and shape of the sacrificial pattern, as shown in FIG. 6.

(52) At **S2033**, the second hard mask layer **123** and the organic sacrificial layer **122** are etched based on the patterned photoresist layer **13**, and the patterned photoresist layer **13** is removed to obtain an organic sacrificial pattern **124**, as shown in FIG. 7.

(53) At **S2034**, a side wall material layer **141** is formed on a side wall and a bottom of the organic sacrificial pattern **124** and a surface of an exposed part of the first hard mask layer **121**, as shown in FIG. 8. The side wall material layer **141** may include, but is not limited to, a silicon oxide layer. The side wall material layer **141** may be formed by using, but not limited to, an atomic layer deposition process.

(54) At **S2035**, the side wall material layer **141** is removed from a top of the organic sacrificial pattern **124** and the surface of the exposed part of the first hard mask layer **121**, and the organic sacrificial pattern **124** is removed to obtain the side wall pattern **14**, as shown in FIG. 9.

(55) At **S2036**, the first hard mask layer **121** and the second part **112** are etched based on the side wall pattern **14**, the side wall pattern **14** and the remaining first hard mask layer **121** are removed to form the isolation groove **113** in the second part **112** to form the sacrificial pattern **1121** by patterning the second part **112**, as shown in FIG. 10.

(56) As an example, **S204** may include the following operations.

(57) At **S2041**, a filling material layer **151** is formed in the isolation groove **113** and on the sacrificial pattern **1121**, as shown in FIG. 11.

(58) At **S2042**, the filling material layer **151** is removed from the sacrificial pattern **1121**, as shown

in FIG. 12.

(59) As an example, an upper surface of the filling layer **15** may be flush with an upper surface of the sacrificial pattern **1121**.

(60) As an example, the filling layer **15** may include a silicon layer or a germanium layer or a silicon germanium layer grown by an epitaxial method.

(61) As an example, after **S20**, the method further includes that the filling layer **15** is subjected with thermal treatment to form a second oxide layer **162** on the surface of the filling layer **15**. As shown in FIG. 14A and FIG. 14B, the second oxide layer **162** and the first oxide layer **161** are seamlessly connected on the surface of the gate trench **103**. Specifically, the silicon layer or germanium layer or silicon germanium layer grown by the epitaxial method may be used as a source or drain of the active area, and the second oxide layer **162** and the first oxide layer **161** may be used as gate oxide layers. The problem of high difficulty in fabrication of forming a gate trench with a relatively large depth-to-width ratio at one time can be avoided by separately fabricating the first trench **1031** and the second trench **1032**.

(62) Referring to **S30** in FIG. 1 and FIG. 15A through FIG. 18B, a gate word line **16** is formed in the gate trench **103** at **S30**.

(63) As an example, the substrate **10** includes an array region and an edge region located at a periphery of the array region, and the gate word line **16** is located in the array region and the edge region. FIGS. 15A, 16A, 17A, and 18A are schematic sectional structural diagrams of the array region, and FIGS. 15B, 16B, 17B, and 18B are schematic sectional structural diagrams of the edge region. The gate word line **16** in the edge region is located in the isolation structure **102**.

(64) As an example, **S30** may include the following operations.

(65) At **S301**, a word line conductive layer **163** is formed in the gate trench **103**. The first trench **1031** is filled up by the word line conductive layer **163** which extends into the second trench **1032**. Specifically, a conductive material layer **1631** may be formed in the gate trench **103** and on the surface of the filling layer **15** first, as shown in FIG. 15A and FIG. 15B. Then, the conductive material layer **1631** is etched back to remove the conductive material layer **1631** on the surface of the filling layer **15** and part of the conductive material layer **1631** located in the second trench **1032**, as shown in FIG. 16A and FIG. 16B.

(66) As an example, the word line conductive layer **163** may include, but is not limited to, a titanium nitride layer. The word line conductive layer **163** may also include a titanium nitride layer and a tungsten layer. In this situation, the titanium nitride layer covers surfaces of the first oxide layer **161** and the second oxide layer **162**. The tungsten layer is located on a surface of the titanium nitride layer away from the first oxide layer **161** and the second oxide layer **162**.

(67) As an example, **S30** further includes the following operations.

(68) At **S302**, an auxiliary conductive layer **164** is formed on an upper surface of the word line conductive layer **163**, as shown in FIG. 17A and FIG. 17B. Specifically, the auxiliary conductive layer **164** may include a polycrystalline silicon layer. In the present embodiment, the auxiliary conductive layer **164** may include a doped polycrystalline silicon layer.

(69) **S30** also includes the following operations.

(70) At **S303**, a dielectric cover layer **165** is formed on the auxiliary conductive layer **164**. The second trench **1032** is filled up by the dielectric cover layer **165** which covers the surface of the filling layer **15**, as shown in FIG. 18A and FIG. 18B. Specifically, the dielectric cover layer **165** may include, but is not limited to, a silicon nitride layer.

(71) As an example, after **S30**, the method further includes the following operations.

(72) At **S50**, a top dielectric layer **17** is formed on the substrate **10**. The top dielectric layer **17** covers the array region and the edge region.

(73) At **S60**, a first interconnection structure **201** is formed in the top dielectric layer **17** on the edge region. The first interconnection structure **201** penetrates through the top dielectric layer **17**, and electrically connects with the word line conductive layer **163**, as shown in FIG. 19.

(74) As an example, the method further includes the following operations between S30 and S50.

(75) At S40, a functional device **18** and a lead-out bonding pad **19** electrically connected to the functional device **18** are formed in the edge region, as shown in FIG. **19**.

(76) As an example, a second interconnection structure **202** is also formed in the top dielectric layer **17** while forming the first interconnection structure **201**. The second interconnection structure **202** is connected to the lead-out bonding pad **19**.

(77) As an example, an upper surface of the auxiliary conductive layer **164** is flush with an upper surface of the lead-out bonding pad **19**. The first interconnection structure **201** is connected to the auxiliary conductive layer **164**. By arranging the auxiliary conductive layer **164**, and making the upper surface of the auxiliary conductive layer **164** flush with the upper surface of the lead-out bonding pad **19**, a height difference between the top of the gate word line **16** and the lead-out bonding pad **19** of the functional device **18** can be eliminated. The first interconnection structure **201** has the same depth as that of the second interconnection structure **202**, which can reduce the scattering of plasmas generated during the etching, thereby facilitating the realization of etching alignment. Therefore, the first interconnection structure **201** will not be offset compared to the gate word line **16**, which can ensure that the first interconnection structure **16** has a sufficiently large contact area with the gate word line, thereby further reducing the contact resistance between the first interconnection structure and the gate word line, and further improving the electrical properties of a storage device to endow the storage device with higher read and write speed and storage efficiency. In this way, the use of a composite gate structure composed of the word line conductive layer **163** and the auxiliary conductive layer **164** is beneficial to improve the gate induced drain leakage (GIDL) effect and further improve the performance of the storage device.

(78) As an example, the top dielectric layer **17** may have a single-layer structure or a stacked structure including a plurality of layers of dielectric layers. In FIG. **19**, it shows the top dielectric layer **17** being a laminated structure including a plurality of layers of dielectric layers. The specific top dielectric layer **17** may include an oxide layer and a silicon nitride layer located on an upper surface of the oxide layer.

(79) Continuing to refer to FIG. **18A**, FIG. **18B**, and FIG. **19** in conjunction with FIG. **2** to FIG. **17B**, the embodiment of the present disclosure further provides a semiconductor structure including a substrate **10**, a gate trench **103**, and a gate word line **16**. The gate trench **103** is located in the substrate **10**, and includes a first trench **1031** and a second trench **1032**. The second trench **1032** is located above the first trench **1031** and communicates with the first trench **1031**. A width of the second trench **1032** is greater than a width of the first trench **1031**. The gate word line **16** is located in the gate trench **103**.

(80) In the embodiment of the present disclosure, the semiconductor structure obtained has the gate word line **16** with a wider upper part by preparing the gate trench **103** with an upper part wider than a lower part. Since the gate word line **16** has the wider upper part, it is convenient to align and contact the subsequently formed first interconnecting structure with the gate word line **16**. Therefore, it can ensure that a sufficiently large contact area can be formed between the first interconnection structure and the gate word line **16**, thereby reducing the contact resistance between the first interconnection structure and the gate word line **16** and improving electrical properties of a storage device to endow the storage device with higher read and write speed and storage efficiency.

(81) As an example, the substrate **10** may include, but is not limited to, a silicon substrate. The substrate **10** includes an active area **101** and an isolation structure **102**.

(82) As an example, the substrate **10** further includes a filling layer **15**, and the second trench **1032** is located in the filling layer **15**.

(83) As an example, a material of the filling layer **15** may include any one of silicon, germanium, or silicon germanium. Specifically, the filling layer **15** can be used as a source or a drain of the active area.

(84) As an example, the gate word line **16** may include: a first oxide layer **161**, a second oxide layer **162** and a word line conductive layer **163**. The first oxide layer **161** is located on a side wall and a bottom of the first trench **1031**. The second oxide layer **162** is located on a side wall of the second trench **1032**. The first trench **1031** is filled up by the word line conductive layer **163** which extends into the second trench **1032**.

(85) As an example, the first oxide layer **161** and the second oxide layer **162** are formed by different processes, and are seamlessly connected on the side wall of the gate trench **103**.

(86) As an example, the word line conductive layer **163** may include, but is not limited to, a titanium nitride layer. The word line conductive layer **163** may also include a titanium nitride layer and a tungsten layer. In this way, the titanium nitride layer covers surfaces of the first oxide layer **161** and the second oxide layer **162**. The tungsten layer is located on the surface of the titanium nitride layer away from the first oxide layer **161** and the second oxide layer **162**.

(87) As an example, an upper surface of the word line conductive layer **163** is located lower than an upper surface of the second trench. The gate word line **16** further includes an auxiliary conductive layer **164**. The auxiliary conductive layer **164** is located on the word line conductive layer **163**, and the auxiliary conductive layer **164** may include a polycrystalline silicon layer. In the present embodiment, the auxiliary conductive layer **164** may include a doped polycrystalline silicon layer. The upper surface of the auxiliary conductive layer **164** is located lower than a top surface of the gate trench **103**.

(88) As an example, the gate word line **163** further includes a dielectric cover layer **165**. The second trench **1032** is filled up by the dielectric cover layer **165** which covers the surface of the filling layer **15**, as shown in FIG. **18A** and FIG. **18B**. Specifically, the dielectric cover layer **165** may include, but is not limited to, a silicon nitride layer.

(89) As an example, the substrate **10** includes an array region and an edge region located at a periphery of the array region. The gate word line **16** is located in the array region and the edge region. The semiconductor structure further includes: a top dielectric layer **17** that covers the array region and the edge region, a first interconnection structure **201** located in the top dielectric layer **17** on the array region, and penetrating through the top dielectric layer **17** to electrically connect with the word line conductive layer **163**.

(90) As an example, the gate word line **16** in the edge region is located in the isolation structure **102**.

(91) As an example, the semiconductor structure further includes a second interconnection structure **202**. The second interconnection structure **202** is located in the top dielectric layer **17** on the edge region, and is connected to the lead-out bonding pad **19**.

(92) As an example, the semiconductor structure further includes a functional device **18** and a lead-out bonding pad **19**. The functional device **18** is located in the edge region, and the lead-out bonding pad **19** is electrically connected to the functional device **18**. The upper surface of the auxiliary conductive layer **164** is flush with the upper surface of the lead-out bonding pad **19**, and is connected to the first interconnection structure **201**. By arranging the auxiliary conductive layer **164**, and making the upper surface of the auxiliary conductive layer **164** flush with the upper surface of the lead-out bonding pad **19**, a height difference between the top of the gate word line **16** and the lead-out bonding pad **19** of the functional device **18** can be eliminated. The first interconnection structure **201** has the same depth as that of the second interconnection structure **202**, which can reduce the scattering of plasmas generated during the etching, thereby facilitating the realization of etching alignment. Therefore, the first interconnection structure **201** will not be offset compared to the gate word line **16**, which can ensure that the first interconnection structure **16** has a sufficiently large contact area with the gate word line, thereby further reducing the contact resistance between the first interconnection structure and the gate word line, and further improving the electrical properties of a storage device to endow storage device with higher read and write speed and storage efficiency. In this way, the use of a composite gate structure composed of the

word line conductive layer **163** and the auxiliary conductive layer **164** is beneficial to improve the gate induced drain leakage (GIDL) effect and further improve the performance of the storage device.

(93) The technical features of the embodiments described above can be arbitrarily combined. In order to make the description concise, all possible combinations of various technical features in the above embodiments are not completely described. However, the combinations of these technical features should be considered as the scope described in the present specification as long as there is no contradiction in them.

(94) The foregoing embodiments represent only a few implementation modes of the present application, and the descriptions are specific and detailed, but should not be construed as limiting the scope of the present application. It should be noted that those of ordinary skill in the art may further make variations and improvements without departing from the conception of the present application, and these variations and improvements all fall within the protection scope of the present application. Therefore, the protection scope of the present application should be subject to the appended claims.

Claims

1. A method for manufacturing a semiconductor structure, comprising: providing a substrate; forming a gate trench in the substrate, the gate trench including a first trench and a second trench, the second trench being located above the first trench, communicating with the first trench, having a width greater than a width of the first trench; and forming a gate word line in the gate trench; wherein the forming the gate trench in the substrate comprises: forming the first trench in the substrate; forming a sacrificial layer, wherein the sacrificial layer comprises a first part and a second part, the first trench is filled up by the first part, and the second part covers an upper surface of the substrate and an upper surface of the first part; forming an isolation groove in the second part to form a sacrificial pattern by patterning the second part, wherein the sacrificial pattern is arranged to correspond to the first part, and has a width greater than a width of the first part; forming a filling layer in the isolation groove, wherein the isolation groove is filled up by with the filling layer; removing the sacrificial pattern to form the second trench; and removing the first part; wherein the filling layer comprises a silicon layer or a germanium layer or a silicon germanium layer grown by an epitaxial method.
2. The method for manufacturing the semiconductor structure of claim 1, further comprising: subjecting the filling layer with thermal treatment to form a second oxide layer on a surface of the filling layer.
3. The method for manufacturing the semiconductor structure of claim 2, further comprising: forming a first oxide layer on a side wall of the first trench, wherein the first oxide layer and the second oxide layer are seamlessly connected on a surface of the gate trench.
4. The method for manufacturing the semiconductor structure of claim 3, wherein the forming the gate word line in the gate trench comprises: forming a word line conductive layer in the gate trench, wherein the first trench is filled up by the word line conductive layer which extends into the second trench.
5. The method for manufacturing the semiconductor structure of claim 1, wherein the substrate comprises an array region and an edge region located at a periphery of the array region; the gate word line is located in the array region and the edge region; the method further comprising: after forming the gate word line, forming a top dielectric layer on the substrate, wherein the top dielectric layer covers the array region and the edge region; forming a first interconnection structure in the top dielectric layer on the edge region, wherein the first interconnection structure penetrates through the top dielectric layer, and electrically connects with the word line conductive layer.

6. The method for manufacturing the semiconductor structure of claim 5, wherein the substrate comprises an active area and an isolation structure; the gate word line in the edge region is located in the isolation structure.

7. The method for manufacturing the semiconductor structure of claim 6, further comprising: forming a functional device and a lead-out bonding pad electrically connected to the functional device in the edge region, prior to the forming the top dielectric layer on the substrate; wherein an upper surface of the word line conductive layer is located lower than the upper surface of the second trench; forming an auxiliary conductive layer on the upper surface of the word line conductive layer, after the forming the word line conductive layer; wherein an upper surface of the auxiliary conductive layer is flush with an upper surface of the lead-out bonding pad, and the first interconnection structure is connected to the auxiliary conductive layer.
